Linear Combination

of u and v

$$c_1 u + c_2 v$$

e.g. $2u - 3v$

if $u = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ $v = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$

and $c_1 u + c_2 v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$$2c_1 + 3c_2 = 1 \qquad c_1 = \tfrac{1}{2} - \tfrac{3}{2}c_2$$

$$3c_1 + 5c_2 = 0$$

$$3\left(\tfrac{1}{2} - \tfrac{3}{2}c_2\right) + 5c_2 = 0$$

$$3/2 - \tfrac{9}{2}c_2 + 5c_2 = 0$$

$$-3/2 = \tfrac{1}{2}c_2 \qquad c_2 = -3$$

$$c_1 = 5$$

$$u \cdot v = \sum_{i=1}^{n} u_i v_i$$

$$(2,3,0,1) \cdot (4,5,6,7)$$
$$= (2 \cdot 4) + (3 \cdot 5) + (0 \cdot 6) + (1 \cdot 7)$$
$$= 8 + 15 + 0 + 7 = 30$$

$$u = (u \cdot e_1) e_1 + (u \cdot e_2) e_2$$

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \frac{\left(\begin{pmatrix} 2 \\ 3 \end{pmatrix} \cdot v\right) v}{v \cdot v} + \frac{\left(\begin{pmatrix} 2 \\ 3 \end{pmatrix} \cdot w\right) w}{w \cdot w}$$

if $v \cdot w = 0$

if $v \cdot w = 0$ $\quad v, w \neq 0$

$v \perp w$

then

$$u = \frac{(u \cdot v)\, v}{v \cdot v} + \frac{(u \cdot w)}{w \cdot w} w$$

in n-dim $\quad v_1, \dots v_n$ orthogonal

$$u = \sum_{i=1}^{n} \frac{(u \cdot v_i)}{v_i \cdot v_i} v_i$$

- if $u_0 \cdot v_0 = 0$

then $z = \dfrac{(z \cdot u_0)}{u_0 \cdot u_0} u_0 + \dfrac{(z \cdot v_0)}{v_0 \cdot v_0} v_0$